NETWORK TRAFFIC ANALYSIS

Wireshark PCAP Forensics Report

A Forensic Examination of traffic.pcapng

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COURSE Ethical Hacking — JOVAC

ASSIGNMENT Assignment 2 — PCAP Analysis

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FILE ANALYZED traffic.pcapng

TOOL USED Wireshark / TShark 4.2.2

1. Objective

This report presents a comprehensive forensic analysis of a real-world network packet capture file (traffic.pcapng) using Wireshark and TShark. The objective is to accurately extract twenty specific data points — covering protocol statistics, host communication patterns, MAC-to-IP mappings, vendor identification, and traffic characteristics — demonstrating practical proficiency in network traffic investigation as required by JOVAC Ethical Hacking Assignment 2.

2. Tools & Environment

Wireshark / TShark 4.2.2	Packet dissection, display filtering, protocol analysis
capinfos 4.2.2	Capture file metadata & duration extraction
Python 3.12	Scripting & report generation
ReportLab (latest)	Professional PDF report generation

3. Capture File Overview

File	traffic.pcapng
Format	pcapng (Wireshark v1.0)
Total Packets	26,456
Duration	348.109916 seconds (~5 min 48 sec)
First Packet	2021-07-26 12:53:50 UTC
Last Packet	2021-07-26 12:59:38 UTC
Avg Packet Rate	75 packets/sec
Capture OS	Windows 10 (build 19042)
Capture Tool	Dumpcap / Wireshark 3.4.7

4. Detailed Question-by-Question Analysis

Each question is answered with the exact TShark/Wireshark display filter used, verified packet counts, and contextual notes derived solely from the packet capture.

E1 Number of DHCP Messages

Filter: `dhcp or bootp`

Answer: **5 Packets**

Notes: *Breakdown: 1 DHCP Inform, 1 Discover, 1 Offer, 1 Request, 1 ACK*

E2 Number of ARP Messages

Filter: `arp`

Answer: **680 Packets**

Notes: *Mix of ARP Requests and Replies across the LAN segment.*

E3 IP Address That Accessed baidu.com

Filter: `dns.qry.name contains "baidu"`

Answer: **10.103.51.159**

Notes: *First DNS query for www.baidu.com appeared in frame 20754. The host also issued HTTP/TLS connections to multiple Baidu subdomains.*

E4 Packets Sent by Source IP 10.103.0.20

Filter: `ip.src == 10.103.0.20`

Answer: **469 Packets**

Notes: *10.103.0.20 acts as the local DNS server / gateway, responding to client queries.*

E5 Number of UDP Packets

Filter: `udp`

Answer: **9,102 Packets**

Notes: *UDP accounts for ~34% of total traffic — primarily DNS, NBNS, and DHCP.*

E6 Number of SMB Packets

Filter: `smb or smb2`

Answer: **518 Packets**

Notes: *SMBv1 traffic (NETLOGON broadcasts) indicates a Windows Active Directory environment.*

E7 Packets Sent to Destination IP 10.103.0.20

Filter: `ip.dst == 10.103.0.20`

Answer: **547 Packets**

Notes: *Clients forward DNS queries and other requests to 10.103.0.20.*

E8 IPv4 Traffic Packet Count

Filter: `ip`

Answer: **24,906 Packets**

Notes: *IPv4 accounts for ~94% of total traffic. IPv6 = 664; ARP/non-IP = ~886.*

E9 Last SMB Packet — Source IP

Filter: `smb or smb2 (last frame)`

Answer: **Frame 26452 — 10.103.50.202**

Notes: *SMB NETLOGON broadcast to 10.103.255.255. Protocol stack: eth→ip→udp→nbdgm→smb.*

E10 MAC Address of IP 151.139.128.14

Filter: `ip.src == 151.139.128.14`

Answer: **00:1c:7f:6c:96:3f**

Notes: *OUI 00:1C:7F belongs to Check Point Software Technologies Ltd. — the local firewall/gateway device.*

E11 Protocol Used in Packet Number 416

Filter: `frame.number == 416`

Answer: **DNS**

Notes: *Protocol stack: eth → ethertype → ip → udp → DNS. UDP port 53. Query from 10.103.51.159 → 10.103.0.20.*

E12 Destination IP in Packet Number 416

Filter: `frame.number == 416`

Answer: **10.103.0.20**

Notes: *DNS query sent to the local DNS server 10.103.0.20 on UDP port 53.*

E13 Source IP of First HTTP Packet

Filter: `http (frame 19007)`

Answer: **10.103.51.159**

Notes: *GET request to ocsp.comodoca.com — an OCSP certificate status check.*

E14 Source Port of First HTTP Packet

Filter: `http → tcp.srcport (frame 19007)`

Answer: **64079**

Notes: *Ephemeral client port 64079 connecting to destination port 80 (HTTP).*

E15 Capture Duration (Seconds)

Filter: `capinfos`

Answer: **348.109916 seconds**

Notes: *Exactly 5 minutes and ~48.11 seconds of traffic captured.*

E16 Number of NBNS Packets

Filter: `nbns`

Answer: **963 Packets**

Notes: *NetBIOS Name Service — typical in Windows LAN environments for hostname resolution.*

E17 TCP Packets with Source Port 443

Filter: `tcp.srcport == 443`

Answer: **7,275 Packets**

Notes: *Inbound HTTPS responses from internet servers to clients in the LAN.*

E18 TCP Packets with Destination Port 443

Filter: `tcp.dstport == 443`

Answer: **5,966 Packets**

Notes: *Outbound HTTPS requests from LAN clients to external HTTPS servers.*

E19 Packets to Destination IP 204.79.197.200

Filter: `ip.dst == 204.79.197.200`

Answer: **116 Packets**

Notes: *204.79.197.200 is part of the Microsoft/Bing CDN infrastructure.*

E20 MAC Address & Vendor of IP 204.79.197.200

Filter: `ip.src == 204.79.197.200`

Answer: **00:1c:7f:6c:96:3f — Check Point**

Notes: *The MAC belongs to the Check Point firewall/gateway — all external traffic routes through it, so external IPs share its MAC in the capture.*

5. Summary Reference Table

Quick-reference table of all 20 questions, the Wireshark filter applied, and the final answer.

#	Question	Wireshark Filter	Answer
E1	Number of DHCP Messages	`dhcp or bootp`	**5 Packets**
E2	Number of ARP Messages	`arp`	**680 Packets**
E3	IP Address That Accessed baidu.com	`dns.qry.name contains "baidu"`	**10.103.51.159**
E4	Packets Sent by Source IP 10.103.0.20	`ip.src == 10.103.0.20`	**469 Packets**
E5	Number of UDP Packets	`udp`	**9,102 Packets**
E6	Number of SMB Packets	`smb or smb2`	**518 Packets**
E7	Packets Sent to Destination IP 10.103.0.20	`ip.dst == 10.103.0.20`	**547 Packets**
E8	IPv4 Traffic Packet Count	`ip`	**24,906 Packets**
E9	Last SMB Packet — Source IP	`smb or smb2 (last frame)`	**Frame 26452 — 10.103.50.202**
E10	MAC Address of IP 151.139.128.14	`ip.src == 151.139.128.14`	**00:1c:7f:6c:96:3f**
E11	Protocol Used in Packet Number 416	`frame.number == 416`	**DNS**
E12	Destination IP in Packet Number 416	`frame.number == 416`	**10.103.0.20**
E13	Source IP of First HTTP Packet	`http (frame 19007)`	**10.103.51.159**
E14	Source Port of First HTTP Packet	`http → tcp.srcport (frame 19007)`	**64079**
E15	Capture Duration (Seconds)	`capinfos`	**348.109916 seconds**
E16	Number of NBNS Packets	`nbns`	**963 Packets**
E17	TCP Packets with Source Port 443	`tcp.srcport == 443`	**7,275 Packets**
E18	TCP Packets with Destination Port 443	`tcp.dstport == 443`	**5,966 Packets**
E19	Packets to Destination IP 204.79.197.200	`ip.dst == 204.79.197.200`	**116 Packets**
E20	MAC Address & Vendor of IP 204.79.197.200	`ip.src == 204.79.197.200`	**00:1c:7f:6c:96:3f — Check Point**

6. Key Security Observations

- **Check Point Firewall Gateway:** MAC address 00:1C:7F:6C:96:3F (Check Point Software Technologies Ltd.) appears as the gateway MAC for all external IP communications, confirming a perimeter firewall is in use.
- **DNS Infrastructure:** IP 10.103.0.20 functions as the internal DNS resolver. All DNS queries — including those for baidu.com, Bing/MSN CDN, and OCSP servers — route through this host.
- **Baidu Browsing Activity:** Host 10.103.51.159 performed extensive browsing of baidu.com and 20+ subdomains (maps, translate, news, drive, video) — indicating deliberate use of the Chinese search engine.
- **HTTPS Traffic Dominance:** TCP port 443 accounts for 7,275 inbound and 5,966 outbound packets, far exceeding plain HTTP. This indicates a modern, encryption-aware user base.
- **Windows Domain Environment:** 518 SMB packets (primarily NETLOGON broadcasts) and 963 NBNS packets confirm an active Windows Active Directory domain with multiple hosts performing NetBIOS name resolution.
- **DHCP Infrastructure:** Only 5 DHCP packets captured — suggesting stable IP assignments with leases not expiring during the 348-second capture window.

7. Conclusion

This analysis successfully extracted all 20 required data points from the traffic.pcapng capture using verified TShark display filters. No answer was guessed or assumed — every result was derived directly from packet-level evidence. The network under analysis is a Windows-based enterprise LAN protected by a Check Point firewall/gateway, with active HTTPS browsing, SMB domain services, DNS resolution, and user activity directed toward both Microsoft and Baidu services. This assignment demonstrates practical command of Wireshark filters and network forensics methodology as required for ethical hacking proficiency.

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